Literature scan: galerkit's a posteriori results against the frontier

18 October 2026. Covers gaps 1–7 of afc_aposteriori_estimators-1.md, with the focus on gap 7 (Smears–Vohralík extended to convection, `sv_estimator`).

1. Answer

The one result that looks new is gap 7. It combines four properties:

- a guaranteed upper bound with no unknown constants;
- ε-robustness in the augmented norm for σ > 0;
- applicability to every conforming algebraically stabilised scheme (AFC, MUAS, SMUAS, BBK), including inexact nonlinear iterates;
- a sketched proof of global robust efficiency in the measure E.

I found no published result that combines these. The closest results each miss at least one:

- **SV20:** reaction–diffusion only.
- **Eigel–Merdon 2016:** stabilised convection–diffusion–reaction (CDR) with guaranteed bounds, but SV20 Prop. 5.1 shows such bounds cannot be robust for reaction dominance.
- **ESV10:** guaranteed, but DG and semi-robust.
- **Verfürth 2005, Tobiska–Verfürth, Cai & Cai (Numer. Math. 2025):** robust, but with unknown constants.
- **AABR13 (SUPG) and ABR17 (AFC):** guaranteed, but in the energy norm and not robust.

The track is right. The publishable core is the efficiency proof, not the construction: replacing f by the convective residual $f^{\sim}$ is a small step that a referee will call obvious. The result becomes clearly stronger if:

- the same estimator is extended uniformly to σ ≥ 0;
- η_S is proved efficient, or replaced by an equilibrated bound of $S^{\sim}$.

Gaps 1–6 are applications or adaptations of known tools. In particular, gap 5's core is Knobloch 2017.

Caveat: the searches covered abstracts and listings. The full texts of the following were not read, and must be checked before any claim:

- Cai & Cai (Numer. Math. 2025);
- the Vohralík–Yousef 2025 chapter;
- AABR13;
- JJK23.

2. Frontier

result	scheme	estimate	hypotheses / limits	mechanism	uniform in
Verfürth, SINUM 43 (2005)	conforming, SUPG	residual estimator ≃ $\|\|\|e\|\|\|$ + $\|b\cdot\nabla e\|_*$ (reliable, globally efficient)	$\|c\|_\infty \le c_b\,\beta$; constants unknown	inf-sup for B in the augmented norm; squeezed bubbles	ε, b, c (lower bound global)
Tobiska–Verfürth, arXiv 1402.5892	SUPG, LPS, subgrid, CIP	same, with a stabilisation consistency error in the upper bound	as above	generic error–residual equivalence	ε, b, c
Cai & Cai, Numer. Math. 2025 (arXiv 2107.06341)	conforming	hybrid explicit estimator, robust in each regime	constants unknown; Verfürth's norm	equivalence to the residual estimator	parameters, per regime
Cheddadi–Fučík–Prieto–Vohralík, M2AN 43 (2009)	conforming, reaction–diffusion	guaranteed; robust via min{equilibrated, residual-type}	reaction–diffusion only	H(div) flux combined with a residual estimate	ε, κ
Smears–Vohralík, M2AN 54 (2020)	conforming, reaction–diffusion, any p	guaranteed, locally efficient	reaction–diffusion only	cut-off-weighted patch problems; weights necessary (Prop. 5.1)	h, ε, κ

result	scheme	estimate	hypotheses / limits	mechanism	uniform in
Ern–Stephansen–Vohralík, JCAM 234 (2010)	DG, CDR	guaranteed; robust in augmented norm + jumps	efficiency constants depend on cut-off ratios	H(div) diffusive and convective reconstructions	semi-robust
Eigel–Merdon, JSC 67 (2016)	conforming SUPG, CDR	guaranteed; augmented norm	not robust for reaction dominance (SV20 Prop. 5.1)	H(div) equilibration	ε, for σ = 0 (numerically)
Ainsworth–Allendes–Barrenechea–Rankin, IJNMF 73 (2013)	SUPG, 3D	guaranteed, energy norm, explicit constants	robustness unverified	equilibrated residuals	? (abstract only)
Allendes–Barrenechea–Rankin, SISC 39 (2017)	AFC / BBK, 2D and 3D	guaranteed energy bound; efficiency up to a stabilisation term	Xu–Zikatanov meshes for the DMP	flux equilibration	not ε-robust
Jha 2021; Ahmed–Jha 2026 (arXiv 2602.14504)	AFC and algebraic stabilisations	residual estimators; numerical study	constants unknown	residual-based	—
galerkit gap 7	any conforming P1 algebraic stabilisation, inexact iterates, 2D/3D	guaranteed for σ > 0; augmented effectivity 3.2–7.3; global efficiency in E (sketch)	σ > 0; efficiency of η_S open	SV20 patches with $\tilde{f}$ and D = εI + A; computable cut-off evaluation; Verfürth bubbles	ε, σ, h (numerically; proof sketched)
Knobloch, LNCSE 120 (2017)	AFC, upwind-type limiter	DMP iff min{a_ij, a_ji} ≤ 0; Limiter 2 gives the DMP on any mesh	—	algebraic	any dimension

3. Open gaps, ranked by novelty-to-effort

1. **Full proof of robust efficiency for the gap 7 estimator.** Written 18 Oct 2026: sv_efficiency_proof.md (local efficiency in local dual norms; global two-sided robustness in E). It rests on four cited lemmas, one of them unverified. The weight-free construction needed a new comparison argument; the proof also corrected the augmented-norm factor to κ = 2 + max{1, c_b}.
   - **What it needs:**
     - SV20 Lemma 4.2 with D = εI + A, which needs Res_A(ψ_a) = 0 (true at the discrete solution);
     - the w = 1 argument;
     - Verfürth's global lower bound;
     - Verfürth's Lemma 3.1, ‖Res‖_* ≲ |||e||| + ‖b·∇e‖_*.
   - **Novelty:** high; this is the paper.
   - **Effort:** medium.
   - **Risk:** the squeezed-bubble estimates for Res_A, whose jumps include A.
2. **One estimator for σ ≥ 0.** Implemented 18 Oct 2026 (`sv_estimator`, see sv_convection_extension.md §9).
   - **What it needs:**
     - at σ = 0 the A-version is a Braess–Schöberl problem, with a mean-zero potential space for interior vertices; compatibility Res_A(ψ_a) = 0 allows it;
     - at inexact iterates, the gap 1 algebraic-error term covers the defect.
   - **Why it matters:** it removes the σ > 0 restriction, the main "only half the problem" objection.
   - **Effort:** low to medium; most of the code exists.
3. **Equilibrate $\tilde{S}$ instead of bounding it by η_S.** Implemented 18 Oct 2026: the equilibrated bound is within 18–30 % of ‖$\tilde{S}$‖_ for σ = 0; efficiency sketch in §9 of the summary.*
   - **What it needs:** treat $\tilde{S}$ as a pure face functional of the jumps of A∇u_h, with an SV20-type patchwise equilibration.
   - **Payoff:** this should give η_S ≲ ‖$\tilde{S}$‖_* and complete the two-sided bound in E.
   - **Novelty:** medium. **Effort:** medium.
4. **Local efficiency in a localisable norm.** Done 18 Oct 2026: sv_efficiency_proof.md §9.
   - Local efficiency holds in the localised measure E_loc = (|||e|||² + Σ_a(‖B_A'e‖²_{,a} + ‖$\tilde{S}$‖²_{,a}))^{1/2}.

- E_loc ≲ E ≲ E_loc + η_{def,S}: the only global term is the computable dual norm of the stabilisation's nodal moments.
- Code: `localized_error`, `eta_Eloc`, `loc_error=True`.
- **The obstacle:** Verfürth's bound is global because ‖b·∇e‖_* does not localise.
- **Novelty:** high. **Effort and risk:** high. Leave this out of the first paper.

5. **3D study (gaps 3, 5, 7).** Implemented 18 Oct 2026: the `sv_sweep` and `sv_adaptive` studies of `guaranteed3d` (144 jobs, `run_sv.py`); see sv_convection_extension.md §12. Smoke runs only; the full runs are pending.
   - **Context:** no 3D numerics of guaranteed estimators for algebraic stabilisations were found beyond ABR17's two examples. JJK23 and Ahmed–Jha 2026 are 2D.
   - **Novelty:** moderate (numerical). **Effort:** low; the full runs only need to be made.
6. **Adaptive stopping of the AFC iteration (gap 1).** Implemented 18 Oct 2026: `EstimatorStop(kind="sv")` with the iteration-free η_alg = κη_{def,A} (`sv_algebraic`); Proposition 5 in sv_efficiency_proof.md §10. [HPSV21] (quasi-optimal cost) needs strong monotonicity, which AFC lacks, so optimal cost for AFC stays open.
   - **Context:** the Ern–Vohralík 2013 framework and later quasi-optimal-cost results were not found applied to AFC.
   - **Verdict:** an application; bundle it into item 1 or 5.

Incremental results; do not claim them as new:

- **Gap 4:** non-symmetric lifting. The symmetric case is in EM16 and BCD04.
- **Gap 6:** the convection analogue of the CFPV09 combination; gap 7 supersedes it.
- **Gap 5:**
  - the condition and the fix are Knobloch 2017;
  - the 3D bisection similarity classes are possibly known in the mesh literature, though no reference was found;
  - the F-BBK weight floor is minor.

Is there a better track? For P1 algebraic stabilisations, no:

- a higher-order, p-robust version (Ern–Vohralík) is irrelevant at P1;
- a residual-based track (Verfürth, Cai & Cai) gives robustness but loses the guarantee, which is the selling point.

The strongest version is gap 7 plus items 2 and 3. Switching costs about one extra lemma (items 2 and 3) and a σ = 0 branch in `sv_estimator`.

4. References

Status tags: [read] full text read; [abstract] abstract or listing only; [unverified] journal details not verified.

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